

Skill-Clustered Network Visualiser — Mathematical Methodology

Correction note — 6 October 2026 (D-77). The table below is headed “real values”, and the introduction speaks of “the current build”. Both mean values actually measured at the time of writing (July 2026), over the 83 generated Phase 1 profiles: $N = 83$, as stated at each point of use. They are not values for the real network, or for the method the project now uses.

Over the real network the cross-algorithm agreement collapsed. ARI was 0.047 and HDBSCAN modularity 0.3352 over 430 people, and 0.0557 and 0.3429 over 442. HDBSCAN was discarded at the Phase 1.5 close-out on 19 July 2026. The mathematics of each object below is unaffected, but the “convergent evidence” the document closes on held for the generated data only. See `docs/phase1_5_methodology.md`. The run that produced these figures survives only in the private repository’s history. The document below is left as written.

A complete reference for every mathematical object in the pipeline, from the embedding space through to graph validation. Written as a standing appendix for the UCL MSc write-up. Reported metrics from the current build are worked in where relevant.

Convergent validation at a glance (real values):

Lens	Metric	Value	What it measures
Geometric separation	Silhouette	0.4745	own-cluster vs nearest-rival distance
Neighbourhood preservation	Trustworthiness	0.9840	honesty of the 2-D projection
Graph community structure	Modularity Q	0.5224	within-community edges vs a random null
Cross-algorithm agreement	ARI	0.8985	greedy graph communities vs HDBSCAN

The pipeline validates a *single* geometric object — the embedding — through four mathematically independent lenses. Their convergence is the core robustness argument.

0. The pipeline as a composition of maps

A person begins as text and is mapped

$$\text{text} \xrightarrow{\text{embed}} x \in \mathbb{R}^{384} \xrightarrow{\text{HDBSCAN}} \text{cluster label} / \xrightarrow{\text{UMAP}} (x, y) \in \mathbb{R}^2 / \xrightarrow{\text{threshold}} \text{graph } G.$$

Everything downstream is derived from the one representation x . The maths splits into (i) the geometry of the embedding space, (ii) the objectives optimised over it, and (iii) the combinatorics of the graph built on top.

1. Embedding space and cosine geometry

Each profile is a vector $x_i \in \mathbb{R}^{384}$ (sentence-transformer `all-MiniLM-L6-v2`). All similarity is **cosine**:

$$\cos \theta_{ij} = \frac{\langle x_i, x_j \rangle}{\|x_i\| \|x_j\|} = \frac{\sum_k x_{ik} x_{jk}}{\sqrt{\sum_k x_{ik}^2} \sqrt{\sum_k x_{jk}^2}}.$$

Cosine discards magnitude and keeps direction, so two profiles with the same skill mix but different text length count as similar. Geometrically this is work on the unit sphere S^{383} . The link to Euclidean distance, for unit vectors:

$$\|\hat{x} - \hat{y}\|^2 = 2 - 2\langle \hat{x}, \hat{y} \rangle = 2(1 - \cos \theta).$$

So cosine distance $d_{\cos} = 1 - \cos \theta$ is monotone in squared Euclidean distance *after normalisation*. This is why using `metric='cosine'` consistently through HDBSCAN and UMAP is coherent — all stages operate on the same spherical geometry.

Scaling note. sklearn’s HDBSCAN computes cosine via brute-force pairwise distances, i.e. $O(n^2)$ in the number of people. Fine at $n = 83$; a documented consideration if the network grows.

2. HDBSCAN — density, a spanning tree, and a persistence hierarchy

The richest stage: estimate density, build a tree, read clusters off its persistence. Parameters used: `min_cluster_size=5, min_samples=3, metric='cosine'`.

Core distance (a k -NN density estimate). With $k = \text{min_samples}$,

$$\text{core}_k(x) = d(x, \text{its } k\text{-th nearest neighbour}).$$

Small core distance $\Rightarrow$ dense neighbourhood (density $\propto k / \text{core}_k(x)^{\dim}$).

Mutual reachability distance (density-aware metric).

$$d_{\text{mreach}}(a, b) = \max\{\text{core}_k(a), \text{core}_k(b), d(a, b)\}.$$

This inflates distances in sparse regions so noise cannot cheaply bridge dense clumps, while leaving dense regions unchanged.

The greedy step (part one — provably optimal). HDBSCAN builds the **minimum spanning tree** of the complete graph weighted by d_{mreach} , sorting edges ascending and merging with union–find. This is **Kruskal’s algorithm**, equivalent to single-linkage clustering on the mutual-reachability metric. *This greedy construction is globally optimal* — see §8 for why.

Condensed hierarchy. Cutting the dendrogram at every height gives a full hierarchy. `min_cluster_size` condenses it: a split is a real bifurcation only if both children have $\geq \text{min_cluster_size}$ points; otherwise the smaller branch is treated as points *falling out* of the parent, not a new cluster.

Stability via persistence. Reparametrise density as $\lambda = 1/\text{distance}$. A cluster C is born at $\lambda_{\text{birth}}(C)$; each member p falls out at λ_p . Cluster stability is the total persistence

$$S(C) = \sum_{p \in C} (\lambda_p - \lambda_{\text{birth}}(C)).$$

Clusters surviving a wide band of density levels score high.

Selection (maximise total stability). HDBSCAN selects the non-overlapping clusters (an antichain in the condensed tree) maximising $\sum S(C)$, resolved bottom-up: keep the parent if $S(\text{parent}) \geq \sum_{\text{children}} S$, else keep the children. Unclaimed points get label -1 (the outliers). This is why HDBSCAN “finds the natural number of clusters” — that number maximises persistence rather than being supplied as k .

Result: 7 clusters, 2 outliers (2.4% outlier rate) over the (83, 384) matrix.

3. Silhouette — geometric separation

For point i in cluster C_I :

$$a(i) = \frac{1}{|C_I| - 1} \sum_{\substack{j \in C_I \\ j \neq i}} d(i, j), \quad b(i) = \min_{J \neq I} \frac{1}{|C_J|} \sum_{j \in C_J} d(i, j),$$

the mean intra-cluster distance and the mean distance to the *nearest rival* cluster. The silhouette coefficient:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \in [-1, 1], \quad s(i) = 0 \text{ for singletons.}$$

The sign answers “is this point better placed here or next door?” The reported **0.4745** is the mean over clustered points; noise (-1) is excluded because it is not a cluster and would otherwise poison $b(i)$ — a documented choice, not an oversight.

4. UMAP — fuzzy simplicial sets and a cross-entropy

A map $\mathbb{R}^{384} \rightarrow \mathbb{R}^2$ preserving neighbourhood structure. Topological in derivation, but the computable core is two probability graphs and a cross-entropy between them. Parameters used: `n_neighbors=15`, `min_dist=0.1`, `metric='cosine'`.

High-dimensional memberships. For each i , let ρ_i be the distance to its nearest neighbour (enforces local connectivity), and pick scale σ_i by binary search so that

$$\sum_j \exp\left(-\frac{\max(0, d(x_i, x_j) - \rho_i)}{\sigma_i}\right) = \log_2 k, \quad k = \text{n_neighbors}.$$

This yields directed weights $p_{j|i}$, symmetrised by a fuzzy union (probabilistic t-conorm):

$$p_{ij} = p_{j|i} + p_{i|j} - p_{j|i} p_{i|j}.$$

Low-dimensional memberships. In the plane,

$$q_{ij} = (1 + a \|y_i - y_j\|^{2b})^{-1},$$

where a, b are fitted (least squares) so the curve matches `min_dist`; for 0.1, roughly $a \approx 1.58$, $b \approx 0.90$. `min_dist` controls how tightly points may clump; `n_neighbors = k` trades local detail against global structure.

Objective (fuzzy-set cross-entropy).

$$\mathcal{L} = \sum_{ij} \left[p_{ij} \log \frac{p_{ij}}{q_{ij}} + (1 - p_{ij}) \log \frac{1 - p_{ij}}{1 - q_{ij}} \right],$$

minimised by SGD with negative sampling. The first term is an *attractive* force along high- p edges; the second a *repulsive* force from sampled non-edges. This attraction/repulsion balance is what pushes clusters apart on the canvas.

Trustworthiness (validation).

$$T(k) = 1 - \frac{2}{n k (2n - 3k - 1)} \sum_i \sum_{j \in U_i^{(k)}} (r_{ij} - k),$$

where $U_i^{(k)}$ are points among i 's k nearest in 2-D but *not* in $\mathbb{R}^{384}$, and r_{ij} is j 's rank from i in the original space. It penalises exactly the false neighbours that make a projection misleading. The reported **0.9840** says the layout is an honest witness — independent of silhouette, which is why it counts as a separate validation.

5. TF-IDF cluster labelling — the lexical layer

Purely a weighting on word counts (no semantics; that is the embedding's job). With sublinear term frequency and smoothed inverse document frequency, on per-person documents (N people):

$$\text{tf}(t, d) = 1 + \log f_{t,d}, \quad \text{idf}(t) = \ln \frac{1 + N}{1 + \text{df}(t)} + 1, \quad w_{t,d} = \text{tf}(t, d) \cdot \text{idf}(t),$$

each document vector L^2 -normalised: $v_d / \|v_d\|$. A cluster's label is the top coordinates of the mean vector

$$\bar{v}_C = \frac{1}{|C|} \sum_{d \in C} v_d.$$

The $\ln$ in idf is load-bearing: as $\text{df}(t) \rightarrow N$, $\text{idf} \rightarrow 0$, so ubiquitous words vanish and only distinctive vocabulary survives. Preprocessing: `ngram_range=(1,2)` (keeps “data science” intact), `min_df=2`, `max_df=0.8`.

Optional refinement (soft-membership weighting). Using the HDBSCAN membership strength π_p (probability) as weights,

$$\bar{v}_C = \frac{\sum_{p \in C} \pi_p v_p}{\sum_{p \in C} \pi_p},$$

down-weights marginal members so labels reflect the cluster core.

6. Graph construction — order statistics and the handshake lemma

From the similarity matrix S (with $S_{ii} = 0$), keep edge (i, j) iff $S_{ij} \geq \tau$, weighted by S_{ij} .

Choosing τ from a target degree. The **handshake lemma**

$$\sum_i \deg(i) = 2m$$

converts a target mean degree d into an edge count $m \approx Nd/2$. Then τ is simply the $(\lceil Nd/2 \rceil)$ -th largest value among the $\binom{N}{2}$ upper-triangular similarities — an **order statistic**. This makes τ reproducible rather than hand-tuned.

Density. density = $\frac{2m}{N(N-1)}$, the fraction of possible edges realised.

Realised values: $N = 83$, target degree 8 $\Rightarrow m = 332$ edges, $\tau = 0.771$, mean degree 8.0, density 0.0976. Isolates (4) and non-trivial component count (13) follow from the strictness of τ : nodes with no peer above 0.771 stay unconnected — deliberate, mirroring HDBSCAN’s treatment of outliers.

7. Modularity — an observed-minus-expected functional

For a weighted graph with adjacency A , node strengths $k_i = \sum_j A_{ij}$, total weight $m = \frac{1}{2} \sum_{ij} A_{ij}$, and a partition $\{c_i\}$:

$$Q = \frac{1}{2m} \sum_{ij} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(c_i, c_j).$$

The term $\frac{k_i k_j}{2m}$ is the expected edge weight between i and j under the **configuration model** — a degree-preserving random graph — so Q measures how much *more* within-community connection exists than chance predicts. Equivalently,

$$Q = \sum_c (e_{cc} - a_c^2),$$

within-community weight fraction minus its expectation. Range $[-\frac{1}{2}, 1)$; $Q > 0.3$ is the usual “structure beyond chance” heuristic.

Result: the HDBSCAN partition scores $Q = 0.5224$ — the clusters are communities in the graph-topological sense too, verified against a null model that knows nothing about the embedding.

8. Greedy community detection — and the deep point

`greedy_modularity_communities` is the **Clauset–Newman–Moore** algorithm:

1. Every node starts in its own community.
2. For each pair of communities, compute the modularity gain ΔQ_{ij} of merging.
3. Perform the merge with the largest positive ΔQ_{ij} .
4. Repeat until no merge increases Q ; return the highest- Q partition seen.

Locally optimal choice, no backtracking — the textbook greedy schema. **Result:** 13 communities, $Q_{\text{greedy}} = 0.5265$ (slightly above HDBSCAN’s, as expected since greedy directly maximises Q).

The greedy paradigm appears twice in this pipeline, with opposite guarantees.

- Inside HDBSCAN (§2), the MST is built greedily (Kruskal) and is **globally optimal**.
- For modularity, greedy CNM is **only a heuristic**: exact modularity maximisation is NP-hard (Brandes et al., 2008), and greedy can settle in a local maximum.

The distinction is structural, not luck. By the **Rado–Edmonds theorem**, a greedy algorithm returns the optimum for *every* weighting precisely when the underlying independent sets form a **matroid**. Spanning forests of a graph form the **graphic matroid**, whose exchange property is exactly what makes Kruskal exact. The space of graph partitions carries no such matroid structure, so greedy loses its guarantee and becomes a fast, usually-good heuristic.

Write-up framing: MST-greedy is exact by matroid theory; modularity-greedy is a heuristic for an NP-hard problem — and the fact that it *still* recovers the HDBSCAN clusters (ARI = 0.8985) is therefore *stronger* evidence, not weaker: an algorithm with no optimality guarantee, working on a different representation (graph topology, not embedding geometry), independently landed on the same structure.

9. Adjusted Rand Index — combinatorics of agreement

Compare the greedy partition U to HDBSCAN’s V via the contingency table $n_{ij} = |U_i \cap V_j|$, with row/column sums a_i, b_j :

$$\text{ARI} = \frac{\sum_{ij} \binom{n_{ij}}{2} - \frac{[\sum_i \binom{a_i}{2}] [\sum_j \binom{b_j}{2}]}{\binom{n}{2}}}{\frac{1}{2} [\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2}] - \frac{[\sum_i \binom{a_i}{2}] [\sum_j \binom{b_j}{2}]}{\binom{n}{2}}}.$$

Each $\binom{n_{ij}}{2}$ counts pairs the two partitions agree to co-cluster; the subtracted term is the expectation under a generalised hypergeometric null (random partitions with fixed block sizes); the denominator normalises so a perfect match gives 1. Hence “corrected for chance”: 0 = agreement no better than random, and negative values are possible. **Result: 0.8985** — high agreement between two representation-independent algorithms.

10. Fruchterman–Reingold spring layout (plot fallback only)

If UMAP coordinates are unavailable, the preview uses a force-directed layout minimising a physical energy with ideal edge length ℓ :

$$f_{\text{attract}}(d) = \frac{d^2}{\ell} \text{ along edges,} \quad f_{\text{repulse}}(d) = \frac{\ell^2}{d} \text{ between all node pairs.}$$

Gradient descent on this energy; not greedy, and presentation-only — it never affects any metric.

11. Metric quick-reference

Symbol	Definition	Range	“Good”	This build
$\cos \theta$	$\langle x, y \rangle / (\ x\ \ y\)$	$[-1, 1]$	$\rightarrow 1$ similar	edge $\tau = 0.771$
$s(i)$	silhouette	$[-1, 1]$	> 0	mean 0.4745
$T(k)$	trustworthiness	$[0, 1]$	$\rightarrow 1$	0.9840
Q	modularity	$[-\frac{1}{2}, 1)$	> 0.3	0.5224 (HDBSCAN), 0.5265 (greedy)
ARI	adjusted Rand	≤ 1	$\rightarrow 1$	0.8985
$S(C)$	HDBSCAN stability	≥ 0	large	(selection criterion)

12. Reference summary

- **Cosine geometry** — work on S^{383} ; consistent metric end-to-end.
- **HDBSCAN** — core distance $\rightarrow$ mutual reachability $\rightarrow$ MST (Kruskal, exact) $\rightarrow$ condensed tree $\rightarrow$ persistence-stability selection.
- **Silhouette / Trustworthiness** — separation and projection-honesty, computed in different spaces, hence independent.
- **UMAP** — fuzzy simplicial sets; cross-entropy between high-/low-dimensional membership graphs.
- **TF-IDF** — lexical distinctiveness via ln-scaled inverse document frequency.
- **Graph** — order-statistic threshold from the handshake lemma; isolates preserved.
- **Modularity** — observed-minus-configuration-model within-community weight.
- **Greedy (CNM)** vs **MST-greedy** — heuristic (NP-hard) vs exact (matroid / Rado-Edmonds); the crux mathematical contrast of the project.
- **ARI** — chance-corrected partition agreement; the cross-algorithm check.

Four independent lenses, one embedding, convergent evidence.